Hardware III
Guided comparative assembly

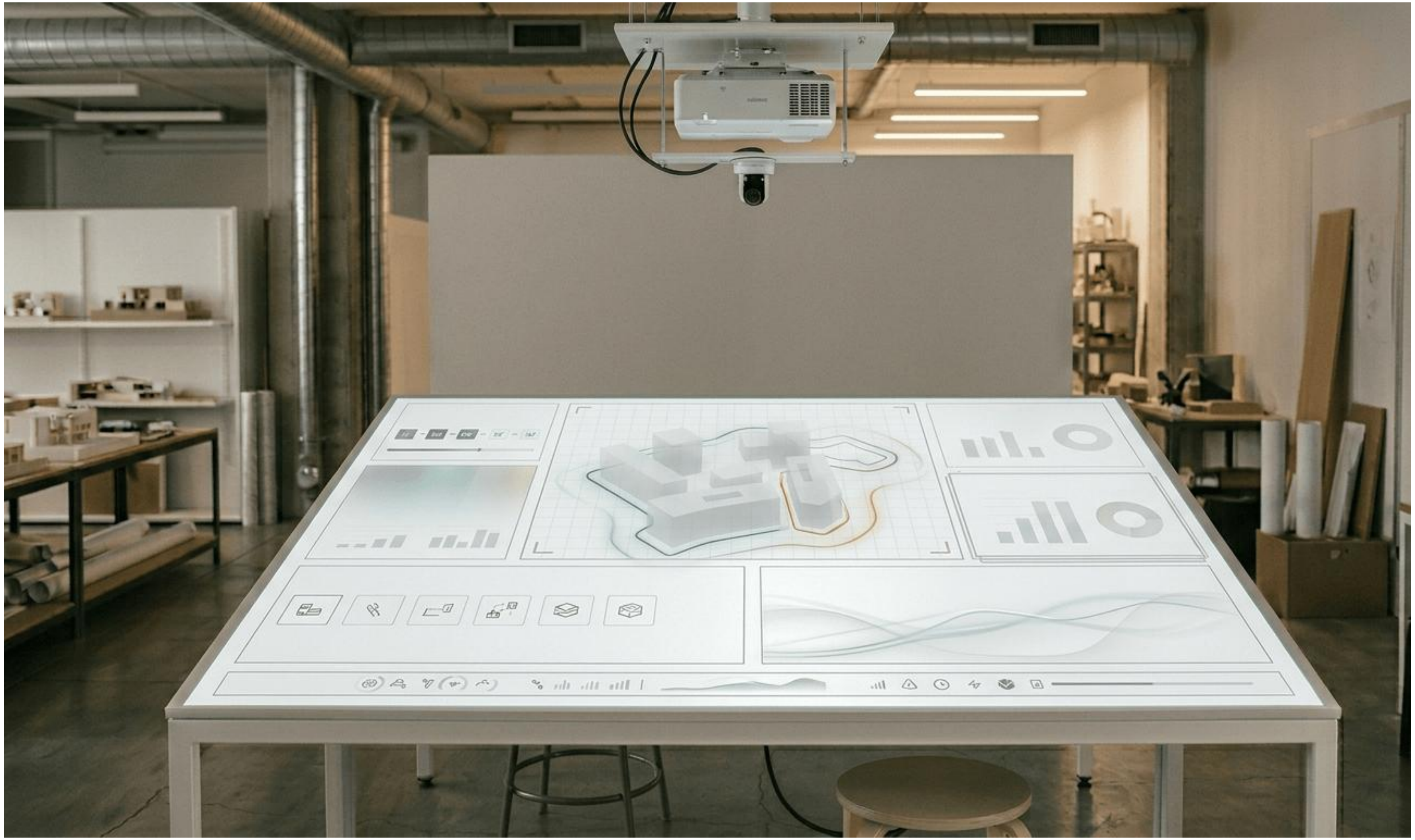
Advanced
Architecture
Barcelona

Construction decisions carry hidden consequences

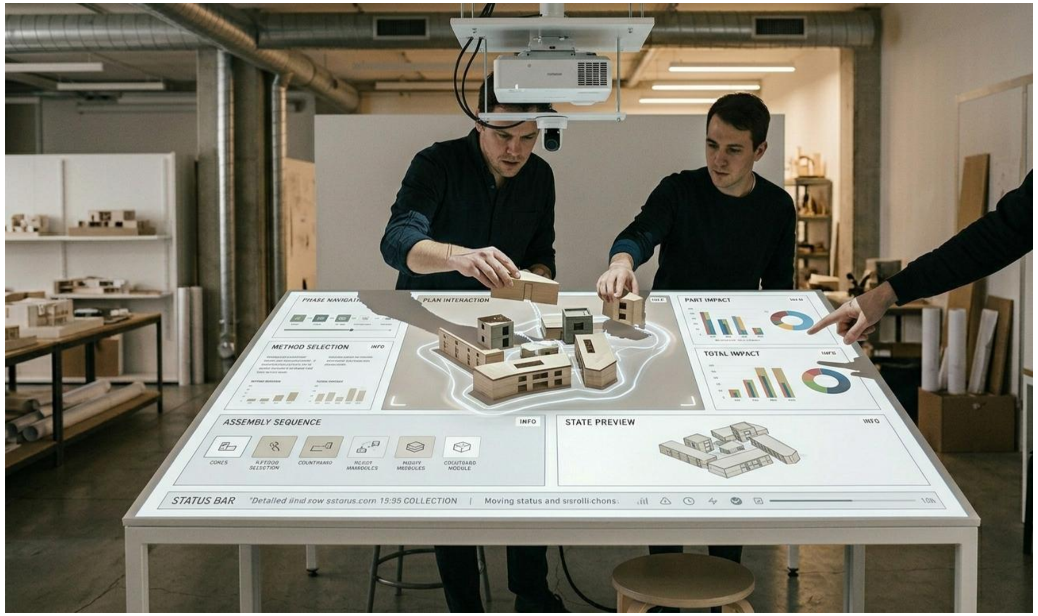
Material selection, construction method, footprint and height are decisions made early in a project, often before environmental or economic data is consulted

For non experts this data is inaccessible. For experts it is contested : methodological assumptions vary and single figure comparisons mislead.

What if you could see those consequences
as you configure the building itself?



A table that reads your building configuration and projects its consequences



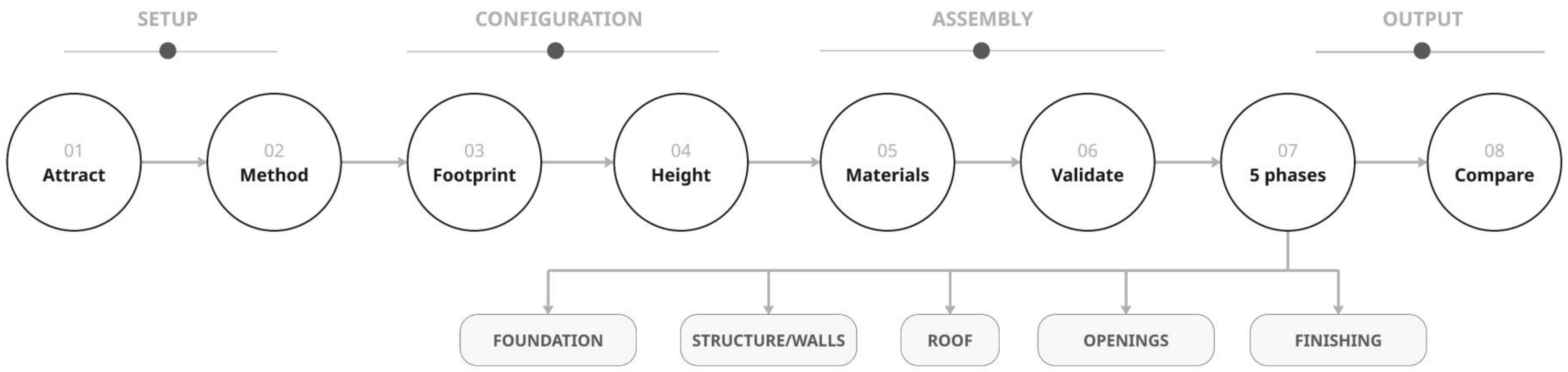
- Select a construction method by placing a physical model on the RFID reader
- Arrange 10 pucks to define your building footprint, projection responds live
- Set height and material with two additional markers
- System walks through 5 construction phases with projected data per phase
- Final comparison view across all configured methods

OUTPUT PER PHASE

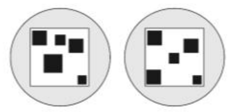
- CO_2 equivalent – Embodied energy
- Labor hours – Construction time
- Cost estimate (2026 EUR)

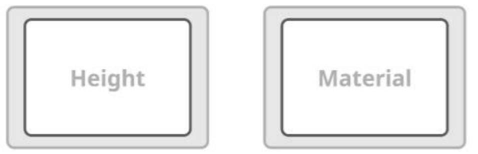
Shown as sourced ranges, not single figures.

8 steps – every transition confirmed by the camera



Objects users interact with :





METHOD MODELS x3

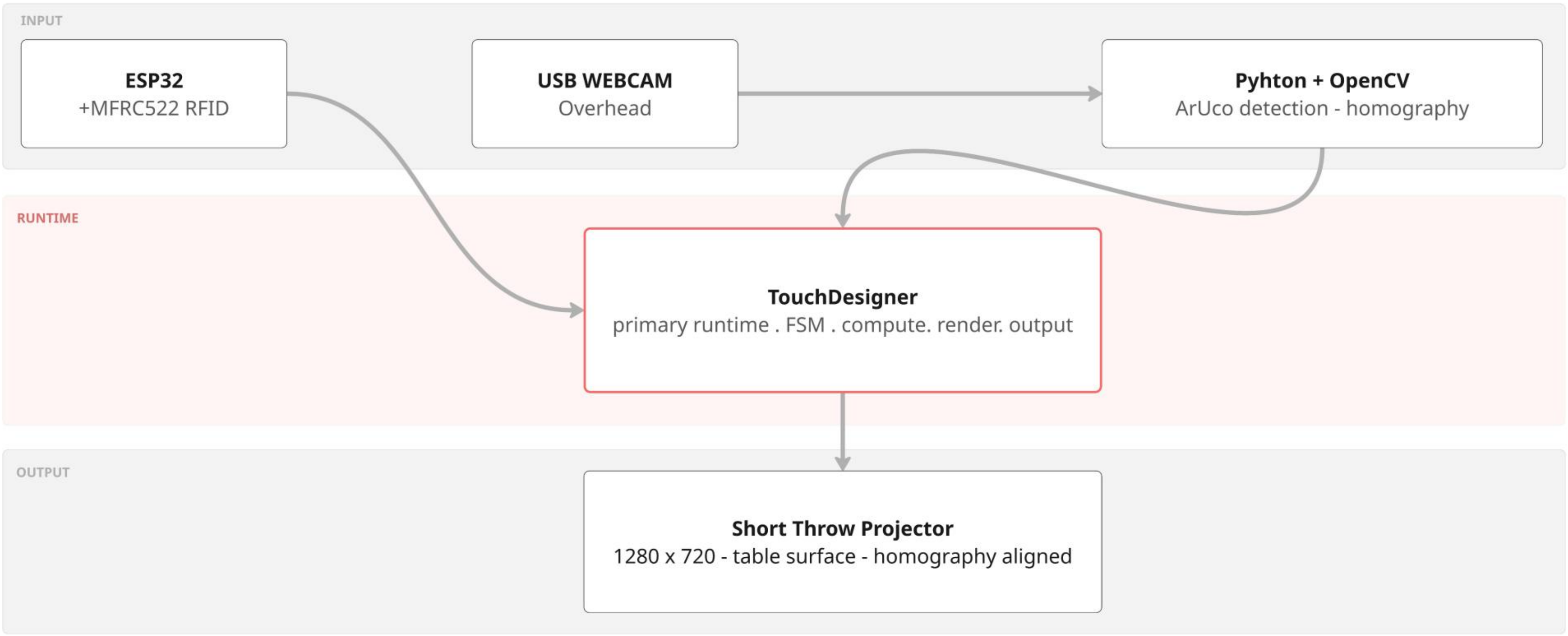
- 3D printed physical representations of Masonry, 3D printed and Prefab construction methods
- Each contains a MIFARE classic 1k RFID tag
- Set height and material with two additional markers
- Placed on RFID pedestal to trigger METHOD state

FOOTPRINT PUCKS x10

- 3D printed pucks, ArUco markers IDs 0 to 9 (4 x 4 dictionary)
- Detected per frame by overhead webcam via openCV
- Convex hull – building footprint area via shoelace formula

CONFIGURATION MARKERS x2

- ArUco ID 10 – height/floor selector
- ArUco ID 11 – material logic selector
- Same detection pipeline as footprint pucks, placement gated by FSM


INPUT
ESP32
+MFRC522 RFID
USB WEBCAM
Overhead
Pyhton + OpenCV
ArUco detection - homography
RUNTIME
TouchDesigner
primary runtime . FSM . compute. render. output
OUTPUT
Short Throw Projector
1280 x 720 - table surface - homography aligned

CONTENT FSM visitor facing 8 states

IDLE attract → **METHOD** RFID select → **FOOTPRINT** 10 pucks → **HEIGHT** floor marker → **MATERIALS** material marker → **VALIDATED** summary → **PHASE N** → **COMPARISON**

WRAPPER Setup & recovery - interrupt content FSM

CALIBRATION_CHECK | ERROR | RESET | MANUAL_OVERRIDE

VISUAL projection output codes - not FSM states

DISCONNECTED | PENDING | INVALID | VALID | IDLE_ANIM | SUMMARY | COMPARISON

COMPUTER VISION

Capture
USB webcam 1080p
overhead

→

ArUco detection
Python openCV 4.8+
4 x 4 dictionary per frame

→

Homography
cam px - projector px
4 calibration YAML - one time setup

→

Grace Period
Frame tracking
10 frames grace period
Timestamp handling - stale data detection

→

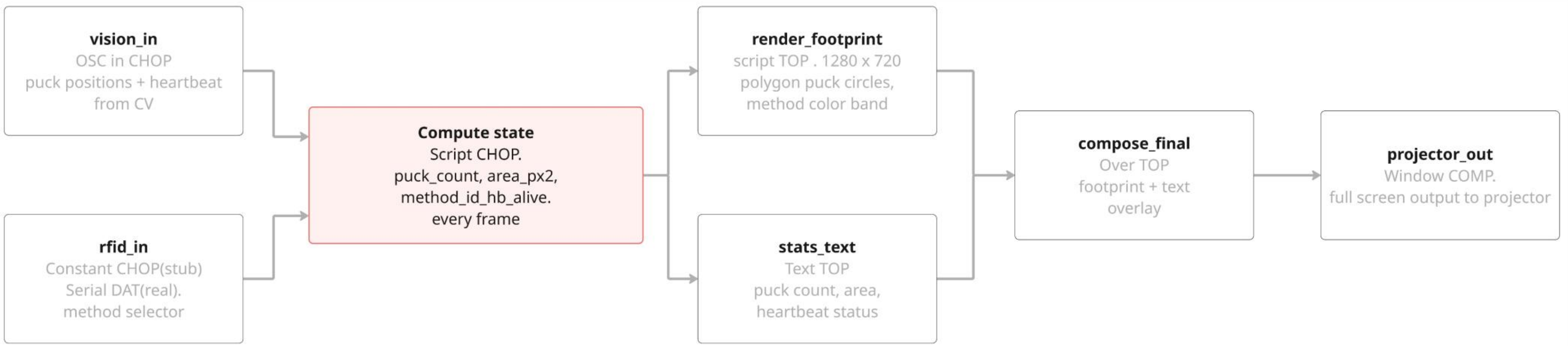
OSC - TouchDesigner
UDP port 7000
/puck/N - /puck/last - /vision/heartbeat

/puck/N i f f
per detected puck : frame number, projector_x, projector_y. sent every frame the puck is visible.

/puck/lost i
sent when a puck leaves the camera frame. Carries the puck ID.

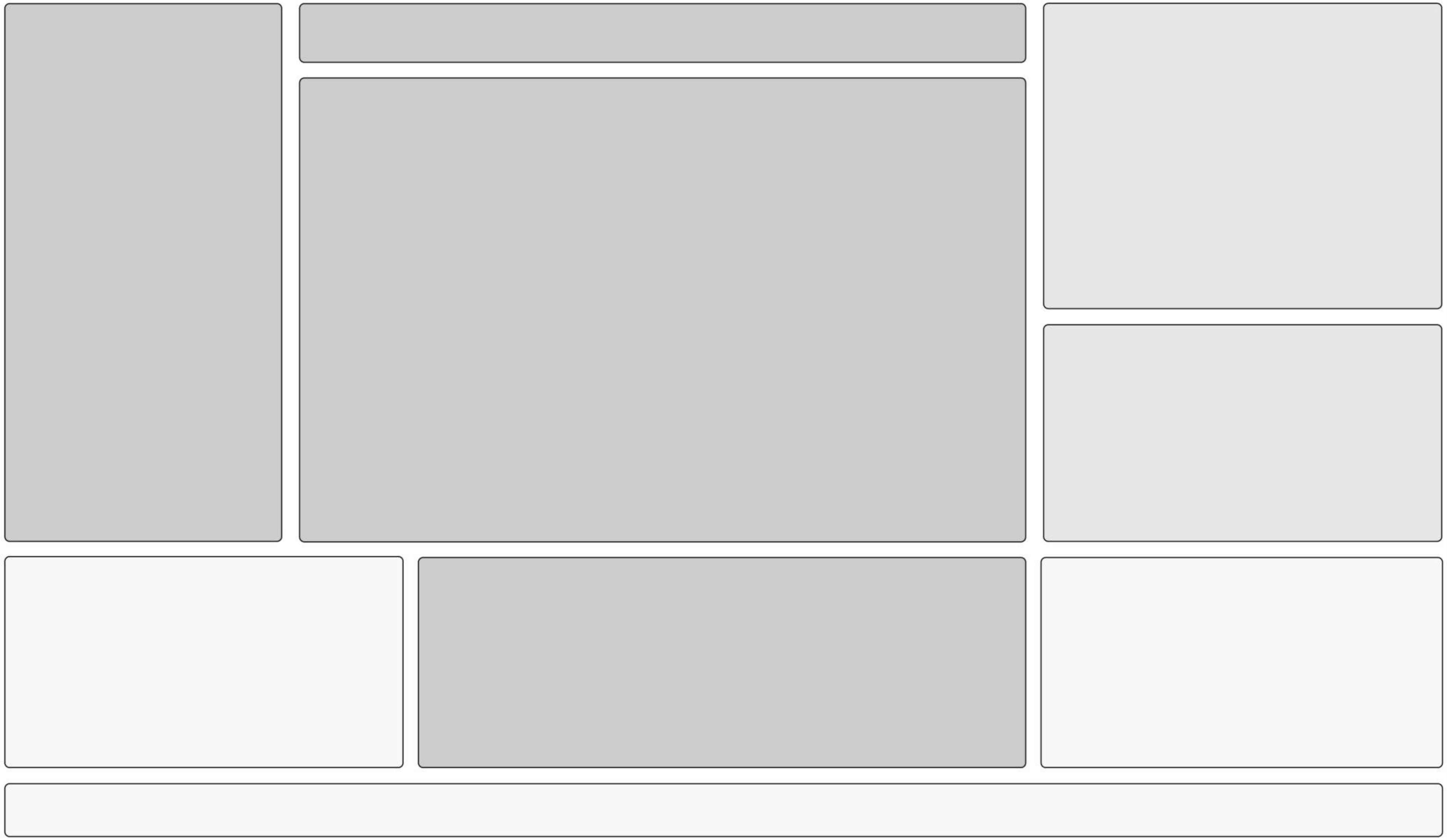
/vision/heartbeat i
Every frame : current frame number. Puck data is considered stale after 10 frames without a heartbeat - triggers ERROR via wrapper FSM.

Seven canonical nodes – the vertical slice network



METHOD	DESCRIPTION	PRIMARY SOURCES	KNOWN WOBBLE	METRIC PER PHASE
Masonry	Traditional fired clay block – Catalonia reference	CYPE . BEDEC (ITeC) . EPD International	Regional variation in clay firing energy intensity	CO2eq . Energy . Labor . Time . cost
3D printed	Concrete extrusion . ICON Apis Cor, COBOD	Journal papers . Vendors EPDs (Tier 2/3)	Cement content range. Geometry efficiency factor	CO2eq . Energy . Labor . Time . cost
Prefab	Modular concrete(MiC) + mass timber CLT	EU JRC . EPFL SXL . Institutional reports	Biogenic carbon accounting . Reuse allocation rules	CO2eq . Energy . Labor . Time . cost

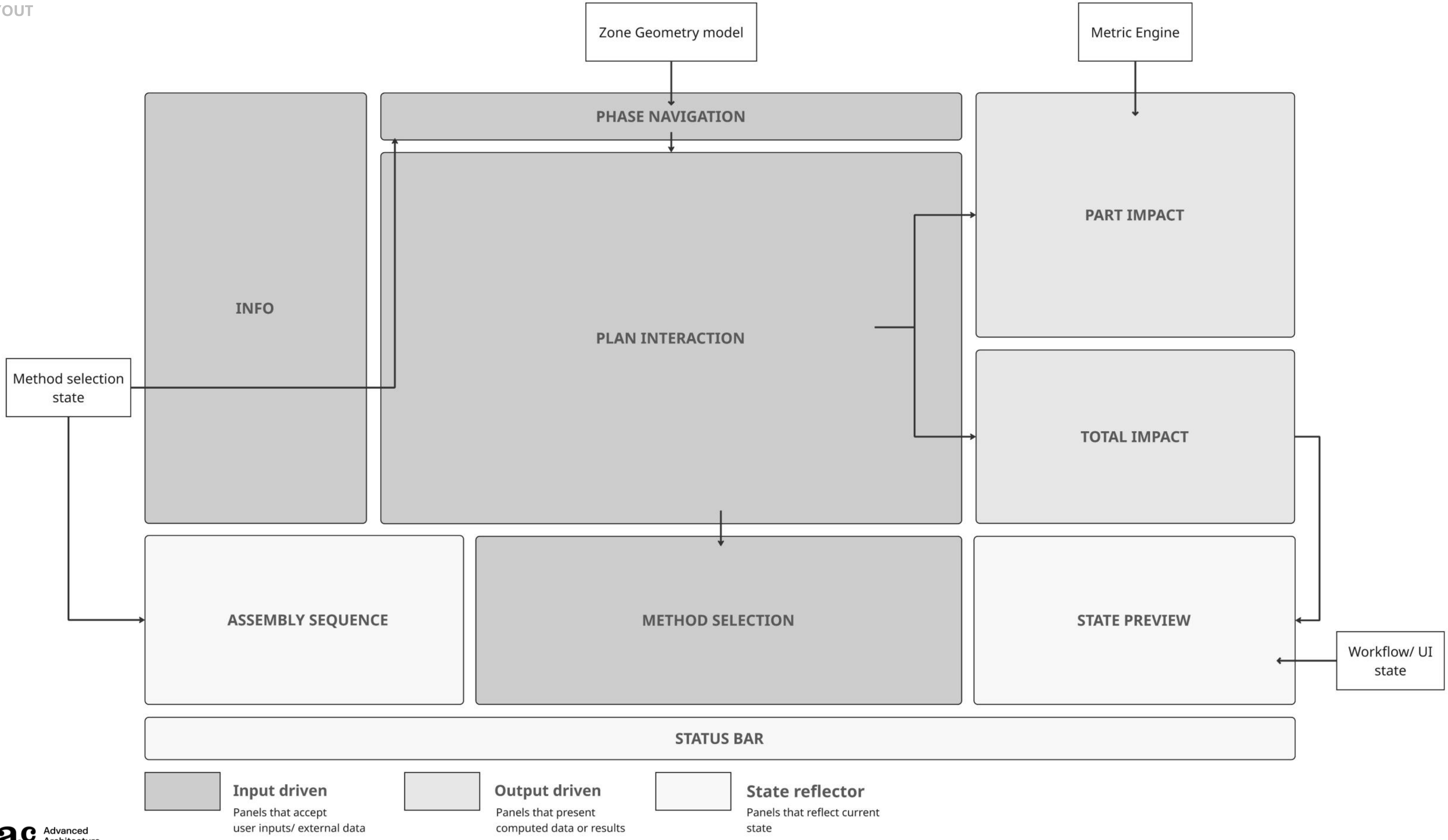
LAYOUT



Input driven
Panels that accept
user inputs/ external data

Output driven
Panels that present
computed data or results

State reflector
Panels that reflect current
state


Zone Geometry model
Metric Engine
PHASE NAVIGATION
INFO
PLAN INTERACTION
PART IMPACT
TOTAL IMPACT
Method selection state
ASSEMBLY SEQUENCE
METHOD SELECTION
STATE PREVIEW
Workflow/ UI state
STATUS BAR
Input driven
Panels that accept user inputs/ external data
Output driven
Panels that present computed data or results
State reflector
Panels that reflect current state